HW1

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Q1

Let X, W, and Z denote matrices.

(a)

If $\mathcal{C}(X) = \mathcal{C}(W)$, show that $P_X = P_W$, i.e., the projection matrices must be the same.

Hint: Consider $(P_W - P_X)'(P_W - P_X)$.

(b)

If X is an $n \times n$ matrix of rank n, show $P_X = I_n$ (i.e., the $n \times n$ identity matrix).

(c)

Let $\mathcal{C}^{\perp}(X)$ denote the orthogonal complement of $\mathcal{C}(X)$ (i.e., $\underline{y} \in \mathcal{C}^{\perp}(X)$ if and only if $\underline{x}'\underline{y} = 0$ for any column $\underline{x}$ of X). If a vector $\underline{b} \in \mathcal{C}(X)$ and $\underline{b} \in \mathcal{C}^{\perp}(X)$, show that $\underline{b}$ must equal the zero vector (this also implies that a non-zero vector in $\mathcal{C}(X)$ must be linearly independent from a non-zero vector in $\mathcal{C}^{\perp}(X)$).

Hint: Write $\underline{b} = X\underline{a}$ *and consider* $\underline{b}'\underline{b}$.

(d)

Let X be an $n \times p$ matrix of rank $r \geq 1$, and Z be an $n \times (n - r)$ matrix with columns formed by $n - r$ linearly independent vectors in $\mathcal{C}^{\perp}(X)$. Show that $P_Z = I_n - P_X$.

Hint: Put r *linearly independent columns of* X *along with the columns of* Z *into one* $n \times n$ *matrix* $\tilde{X}$ *(if* $n = r$*, technically* Z *is just a zero vector). Compute* $P_{\tilde{X}}$ *and use (a)-(b).*

(e)

For a vector $\underline{y}$, show that $P_X\underline{y} = 0$ if and only if $\underline{y} \in \mathcal{C}^{\perp}(X)$.

Hint: use (d) and multiply $I = I - P_X + P_X$ *by* $\underline{y}$

(f)

Show that $P_X + P_W$ is a projection matrix if and only if $X'W = 0$ (the columns of X and W must be orthogonal).

Hint: If $P_X + P_W$ is a projection matrix, one can show $P_X P_W = -P_W P_X$ must hold; this relation implies that any vector $\underline{y}$ that is in both $\mathcal{C}(X)$ and $\mathcal{C}(W)$ must be zero (why?). Further, for $M = P_X P_W$, the relation also gives $M'M = P_X P_W P_W$ and $M'M = P_W P_X P_W$, meaning any column of $M'M$ belongs to $\mathcal{C}^{\perp}(X)$ and to $\mathcal{C}^{\perp}(W)$.

(g)

Show that $P_X - P_W$ is a projection matrix if and only if $\mathcal{C}(W) \subset \mathcal{C}(X)$.

Hint: If $P_X - P_W$ is a projection matrix then $(I - P_X)P_W = -P_W(I - P_X)$ must hold; then the proof is kind of the same as in (f), using that (from (d)) $P_Z = I_n - P_X$ holds, where Z is a matrix with linearly independent columns that span $\mathcal{C}^{\perp}(X)$.

2.

Suppose that $\mathcal{C}(C)$ is a subspace of $\mathcal{C}(A)$ and that $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(A)$.

(a)

Show that $AA^-C = C$ and $DA^-A = D$.

(b)

Using (a), show that

$$\begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix} \begin{bmatrix} A & C \\ D & B \end{bmatrix} \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix},$$

where $Q = B - DA^-C$, and I denotes the identity matrix of appropriate dimensions.

(c)

Using (b) with the fact that the two matrices defined by "I on the diagonals" are square and invertible in (b), show that

$$\begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix}$$

is a generalized inverse of

$$M \equiv \begin{bmatrix} A & C \\ D & B \end{bmatrix}.$$

(d)

Show that the above generalized inverse matches that given in (i) of Theorem 1.4.

(e)

Show that the conditions imposed on the matrices B, C, and D in part (ii) of Theorem 1.4 are satisfied if there exists a matrix $Z = (Z_1, Z_2)$ for which

$$M = Z'Z = (Z_1, Z_2)'(Z_1, Z_2) = \begin{bmatrix} Z_1'Z_1 & Z_1'Z_2 \\ Z_2'Z_1 & Z_2'Z_2 \end{bmatrix} = \begin{bmatrix} A & C \\ D & B \end{bmatrix}.$$

3.

Note to self, see:

- Dale L. Zimmerman, Linear Model Theory: Exercises and Solutions (Springer, 2020), Chapter 3, "Generalized Inverses and Solutions to Systems of Linear Equations"
- https://people.stat.sc.edu/Tebbs/stat714/f10notes.pdf

For each of the following statements, indicate whether the statement is true or false, and explain.

(a)

If $Ax = y$ has a unique solution, then A must be a nonsingular matrix.

False. Let A be an $n \times m$ matrix and suppose $A\mathbf{x} = \mathbf{y}$ is consistent, with particular solution $\mathbf{x}_0$. The full solution set is $\mathbf{x}_0 + \mathcal{N}(A)$, so the solution is unique if and only if $\mathcal{N}(A) = \{\mathbf{0}\}$, i.e., A has full column rank. This is purely a rank condition and has nothing intrinsically to do with A being square — indeed, "nonsingular" is a term defined only for square matrices. As a counterexample, let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix},$$

a 3×2 matrix of full column rank. For any $\mathbf{y}$ in the column space of A — say $\mathbf{y} = A(2, -1)^\top$ — the system $A\mathbf{x} = \mathbf{y}$ has the unique solution $\mathbf{x} = (2, -1)^\top$, even though A is not square and so cannot properly be called nonsingular at all.

(b)

If G is a generalized inverse of A, then $GA = I$, where I denotes the identity matrix of appropriate dimensions.

False. The defining property of a generalized inverse is only $AGA = A$; nothing in this equation forces $GA = I$. As a counterexample, let $A = \text{diag}(1, 0)$, a symmetric, singular, idempotent matrix ($A^2 = A$), and take $G = I$. Then $AGA = A \cdot I \cdot A = A^2 = A$, so $G = I$ is a valid generalized inverse of A. However, $GA = IA = A \neq I$, since A is singular. This single example is enough to show that a valid generalized inverse need not satisfy $GA = I$.

(c)

If A has a generalized inverse G such that $GA = I$, then A must be a (square) nonsingular matrix.

False. The condition $GA = I_m$ forces A (of size $n \times m$) to have full column rank, but this is again a rank condition, not a requirement that $n = m$. Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix},$$

a 3×2 matrix of full column rank, so that $A^\top A$ is 2×2 and nonsingular. Setting $G = (A^\top A)^{-1} A^\top$ gives $GA = (A^\top A)^{-1} A^\top A = I_2$ and $AGA = A(A^\top A)^{-1} A^\top A = A$, so G is a genuine generalized inverse of A satisfying $GA = I$. Yet A itself is 3×2 — rectangular, not square — and therefore cannot be nonsingular. (This G is, in fact, the Moore–Penrose pseudoinverse of A in the full-column-rank case, the same object used to prove part (e).)

(d)

If A is a symmetric matrix, then all its generalized inverses are also symmetric.

False. Symmetry of A constrains A, not G: the equation $AGA = A$ generally admits many solutions G, most of which are not symmetric. Let $A = \operatorname{diag}(1, 0)$, a symmetric matrix, and take

$$G = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Direct computation gives $AGA = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = A$, so G is a valid generalized inverse of A, but $G \neq G^\top$. This counterexample shows that not every generalized inverse of a symmetric matrix is itself symmetric.

(e)

If A is a symmetric matrix, then it has at least one symmetric generalized inverse.

True. Unlike the previous parts, this is an existence claim, so a counterexample cannot settle it — a constructive proof is required instead. Because A is symmetric, the spectral theorem guarantees an orthogonal matrix P (with $P^\top P = PP^\top = I$) and a diagonal matrix $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ of eigenvalues such that $A = PDP^\top$. Define D^- to be the diagonal matrix whose ith entry is $1/\lambda_i$ when $\lambda_i \neq 0$ and 0 when $\lambda_i = 0$, and set $G = PD^-P^\top$. Since D^- is diagonal, $G^\top = P(D^-)^\top P^\top = PD^-P^\top = G$, so G is symmetric. Moreover, since $P^\top P = I$ and $DD^-D = D$ entrywise regardless of whether λ_i is zero,

$$AGA = (PDP^\top)(PD^-P^\top)(PDP^\top) = P(DD^-D)P^\top = PDP^\top = A.$$

Thus $G = PD^-P^\top$ is symmetric and satisfies $AGA = A$, proving that every symmetric matrix has at least one symmetric generalized inverse.

4.

For each of the following questions, give, if possible, a 3×3 **nondiagonal** matrix A that has the stated property.

(a)

A is a generalized inverse of A.

Possible. The condition A is a generalized inverse of itself means $A \cdot A \cdot A = A$, i.e. $A^3 = A$. This holds for any idempotent matrix, since $A^2 = A$ implies $A^3 = A^2 \cdot A = A \cdot A = A^2 = A$. Take

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

a nondiagonal matrix with $A^2 = A$. Consequently $A^3 = A$, so A is a generalized inverse of itself.

(b)

A' is a generalized inverse of A.

Possible. We need $AA^\top A = A$, which holds whenever $A^\top = A^{-1}$, i.e. whenever A is orthogonal. A nondiagonal orthogonal matrix is easy to produce: let A be the 3×3 cyclic permutation matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Since $A^\top A = I$, we have $A^\top = A^{-1}$, and so $AA^\top A = AA^{-1}A = A$. Note $A^\top \neq A$ here (the transpose is the inverse permutation), which nicely illustrates that A' need not equal A to serve as its generalized inverse.

(c)

The identity matrix I is a generalized inverse of A.

Possible. The condition is $AIA = A^2 = A$, i.e. A is idempotent — exactly the same requirement used in part (a). The matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

from part (a) again works, since $A^2 = A$ directly gives $AIA = A$. It's worth noting that idempotency ($A^2 = A$) is strictly stronger than the condition $A^3 = A$ needed in (a): every idempotent matrix satisfies (a), but not every matrix satisfying (a) is idempotent (a nondiagonal reflection, with $A^2 = I \neq A$ but $A^3 = A$, would satisfy (a) without satisfying (c)).

(d)

Every 3×3 matrix is a generalized inverse of A.

Not possible. Requiring $AGA = A$ for *every* 3×3 matrix G is an extremely strong condition. Taking $G = 0$ in particular forces $A \cdot 0 \cdot A = 0 = A$, so A must be the zero matrix; and conversely, if $A = 0$ then $AGA = 0 = A$ trivially holds for any G. Hence $A = 0$ is the *unique* matrix with this property. But the zero matrix has all off-diagonal entries equal to zero, so it is itself a diagonal matrix — there is no nondiagonal 3×3 matrix for which every matrix serves as a generalized inverse.

(e)

Every generalized inverse of A is nonsingular.

Possible. This property holds if and only if A is nonsingular. If A is nonsingular, the equation $AGA = A$ has the unique solution $G = A^{-1}$ (multiply on the left and right by A^{-1}), and A^{-1} is of course nonsingular. Conversely, if A is singular with $\operatorname{rank}(A) = r < 3$, a reflexive generalized inverse always exists — one additionally satisfying $GAG = G$ — and any such G necessarily has $\operatorname{rank}(G) = \operatorname{rank}(A) = r < 3$, making it singular; so a singular A always admits at least one singular generalized inverse, ruling it out. A nondiagonal nonsingular example:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det(A) = 1 \neq 0.$$

Its unique generalized inverse, A^{-1}, is nonsingular by construction.

5.

For any matrix A, let B and C be any two matrices that satisfy $A'AB = A'$ and $AA'C' = A$ (the existence of such B and C follows from Corollary 2 to Lemma 1.4). Let $G = CAB$. Show that G satisfies the following four conditions:

From lecture notes, we have:

Preliminary facts. Corollary 2 to Lemma 1.4 (with $X = A$) guarantees the matrix B satisfying $A'AB = A'$, and Corollary 1 to Lemma 1.5 (again $X = A$) immediately gives

$$ABA = B'A'A = A. \qquad (\star)$$

Corollary 5 to Lemma 1.5 identifies AB as the orthogonal projection matrix onto $C(A)$, which by Corollary 4 to Lemma 1.5 is symmetric:

$$(AB)' = AB. \qquad (\star\star)$$

Applying the *same two corollaries* with $X = A'$ in place of $X = A$ produces the C-side facts. Here $X'X = AA'$ and $X' = A$, so the condition $AA'C' = A$ is exactly Corollary 2 to Lemma 1.4's existence statement for a "B" attached to $X = A'$; Corollary 1 to Lemma 1.5 then gives $A'C'A' = A'$, and transposing (using $(PQR)' = R'Q'P'$) turns this into

$$ACA = A. \qquad (\star\star\star)$$

Likewise, Corollary 4 to Lemma 1.5 says $A'C'$ is symmetric, i.e. $(A'C')' = A'C'$; since $(A'C')' = CA$, this reads $CA = A'C'$, and transposing once more gives

$$(CA)' = A'C' = CA. \qquad (\star\star\star\star)$$

So the notes hand us, at no extra cost, four identities: $ABA = A$, $ACA = A$, AB symmetric, and CA symmetric — everything needed for (i)–(iv) below.

(i)

$AGA = A$,

Substituting $G = CAB$ and grouping,

$$AGA = A(CAB)A = (ACA)(BA) = A(BA) = ABA = A,$$

using $(\star\star\star)$ to collapse $ACA \to A$, then $(\star)$ to collapse $ABA \to A$.

(ii)

$GAG = G$,

First note the two identities that also drive (iii) and (iv): $GA = (CAB)A = C(ABA) = CA$ by $(\star)$, and $AG = A(CAB) = (ACA)B = AB$ by $(\star\star\star)$. Using $GA = CA$,

$$GAG = (GA)G = (CA)(CAB) = C(ACA)B = CAB = G,$$

again invoking $(\star\star\star)$ to collapse the middle ACA to A.

(iii)

$(GA)' = GA$, and

Since $GA = CA$ (shown in (ii)), symmetry of GA is exactly ($\star\star\star\star$):

$$(GA)' = (CA)' = CA = GA.$$

(iv)

$(AG)' = AG$.

Since $AG = AB$ (shown in (ii)), symmetry of AG is exactly ($\star\star$):

$$(AG)' = (AB)' = AB = AG.$$